

LEVEL 7 · CHALLENGE

First Contact

I. The measurement

Along the sight-line at galactic longitude $\ell = 30^\circ$, in the plane of the Galaxy, the 21-centimetre line of neutral hydrogen is received. The component at the sub-central point shows its line shifted from the rest frequency $f_0 = 1.420$ GHz by $\Delta f = -521$ kHz. The Sun travels a circular orbit of speed $v_\odot = 220$ km/s at radius $R_\odot = 8.00$ kpc. Treat the gas as circular and the rotation flat.

Carry the one measurement forward; each result feeds the next:

- the line-of-sight speed of the sub-central gas;
- the Galaxy's circular speed at the sub-central point;
- the mass enclosed within the sub-central radius, in 10^9 solar masses;
- the period of one circular orbit at that radius, in millions of years.

Round each to a whole number; written in order, they are the number that drives the machine.

II. The machine

generator.py scrambled transmission.bin. It is given in full; the driving number is not. It runs forward. Running it again will not give the message back — you must work out how to undo what it does. Recover the 1679 bits, and remember that 1679 was not an arbitrary length.

III. The picture, and the star

What appears carries figures, coordinates, and a name. The coordinates say where the call was received. The name says where to look next — a star. Identify it.

IV. The outer dark

At the Galaxy's edge the stars orbit faster than their visible mass allows; MOND explains this without dark matter. In the deep-MOND regime ($a \ll a_0$), a body orbiting an enclosed mass M settles to a flat circular speed. Derive that speed and compute it, in whole km/s, using the mass you found in step I and $a_0 = 1.2 \times 10^{-10}$ m/s².

That number is a key. The waypoint below is XORed with it (as repeating ASCII digits); decode it to find where the last fragment waits:

```
59424041450e1e194450454054545d5f18575e5b1b06525c0866476950
```

What waits there was sealed with a short word. You already know how it begins.

Files	transmission.bin, generator.py
-------	--------------------------------

Constants	$1 \text{ kpc} = 3.09 \times 10^{19} \text{ m}$ $1 M_{\odot} = 1.99 \times 10^{30} \text{ kg}$ $G = 6.674 \times 10^{-11}$ $c = 3.00 \times 10^5 \text{ km/s}$
Flag	CTC{...}